MARIA YOHANA VIANNY LEO

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SUMMARY

Computer Science undergraduate specializing in Data Science at BINUS University (GPA: 3.85/4.00), with hands-on experience in data analytics, machine learning, and predictive modeling using Python, Pandas, and scikit-learn. Experienced in data preprocessing, exploratory data analysis, model evaluation, and deploying data-driven applications. Strong analytical thinking, problem-solving, leadership, and teamwork skills demonstrated through technical projects, research, and organizational involvement.

EDUCATION

BINUS University *Jakarta*
Bachelor of Computer Science | Data Science Streaming Program *2024 – 2028*
GPA: 3.85/ 4.00
Relevant coursework : Artificial Intelligence, Database Technology, Data Mining and Visualization, Big Data Infrastructure and Technology, Software Engineering, Machine Learning, Deep Learning, Text Mining

SMAN1 Larantuka *Larantuka*
Mathematics and Natural Sciences Major *2021 – 2024*

PUBLICATIONS

"Scholarship Eligibility Profiling Using K-Means Clustering and Comparative Machine Learning" (ICTIM 2026)
1st Author | Accepted for Publication

August 2026

- Developed a two-stage machine learning framework using 10,798 PISA 2022 Brazil student records to generate scholarship eligibility profiles from academic, socioeconomic, and ICT access indicators. Applied K-Means clustering for pseudo-label generation and compared five classification algorithms, with SVM achieving 99.0% accuracy, 0.983 Macro F1-score, and 0.985 mean Macro F1 in five-fold cross-validation. Conducted clustering validation, feature ablation, permutation importance, leakage analysis, and robustness testing to evaluate model reliability and interpretability.

PROJECTS

Movie Recommender AI | Hybrid Content-Based Movie Recommendation System *Oct 2026*
Live App | GitHub

- Developed a content-based movie recommendation system on the TMDB 5000 Movie dataset (~4,800 films), combining Sentence Transformer (all-MiniLM-L6-v2) synopsis embeddings with genre and keyword similarity in a 60/25/15 weighted score to return the top 5 most similar films.
- Built a FastAPI backend that serves pre-computed similarity scores, with the model files stored on Hugging Face.
- Deployed a responsive Next.js frontend on Vercel featuring case-insensitive, typo-tolerant title search, 3D movie cards, and detail pop-ups showing synopsis, rating, runtime, and budget.

Hotel Booking Cancellation Analysis | Data Analysis & Dashboard *June 2026*
Dashboard | GitHub

- Analyzed 35,639 hotel booking records across 31 columns with Python (Pandas) to uncover the factors behind cancellations, identifying an overall cancellation rate of 32.5%.
- Found that lead time, seasonality, price, and customer intent are strong drivers: long-lead-time bookings canceled at 42.0%, July peaked at 45.1%, and cancellations dropped from 43.2% to 14.7% for bookings with 2 special requests.
- Built an interactive Excel dashboard and a React web dashboard to present the cancellation drivers and key findings visually.

Student CGPA Prediction | Machine Learning Regression (Group Project) *May 2026*
GitHub

- Built a leakage-free scikit-learn pipeline, fitted on training data only, to predict student CGPA from study habits and social conditions (1,194 survey records, 22 features), deliberately excluding prior

academic grades.

- Compared Linear, Lasso, Ridge, and Random Forest models with 5-fold cross-validation; the tuned Random Forest performed best (R^2 0.291, MAE 0.443), with current semester, class attendance, and social media hours as the top drivers.
- Deployed a Streamlit prediction app with 22 inputs that returns a CGPA estimate, a performance category, and a recommendation.

Green Pulse | AI-Based Smart Irrigation System *Nov 2025*

Live App | GitHub

- Developed an AI-based watering decision support system that classifies indoor plants as Healthy or Needs Watering from soil moisture, temperature, and humidity, using 3,000 sensor records from a Mendeley Data dataset.
- Benchmarked Logistic Regression, Decision Tree, Random Forest, and SVM with a stratified 80/20 split and 5-fold cross-validated balanced accuracy; selected Logistic Regression for stable performance and interpretability (~99.8% test accuracy, ~99.9% CV balanced accuracy).
- Deployed the model through FastAPI into a responsive web application supporting multi-plant management, simulated sensor input, prediction history, and watering notifications, with future scope for physical IoT sensor integration.

EXPERIENCE

STEMpedia | Codeavour 7.0 International Showdown

International Event Operations Associate *May 2026*

- Facilitated technical setups, participant registrations, competition task executions, and crowd management for an international robotics and AI event at BINUS University Alam Sutera.
- Coordinated operational support and cross-cultural communication for student competitors hailing from various countries, ensuring smooth tournament execution.
- Awarded an official **Letter of Recommendation from the FSEF Director** for outstanding leadership, operational agility, and fast-paced problem-solving.

STAMANARA (Dance Organization) | BINUS University

Human Resource Development (HRD) *Feb 2025 – Feb 2026*

- Led member recruitment campaigns, conducted structured candidate interviews, and evaluated applicant suitability to build a strong cohort.
- Maintained participant records, attendance tracking, and communication channels for major programs and alumni networks, and handled registration and orientation information at organization booths.

Event Division | Cross-Institutional Benchmarking *Oct 2025 – Jan 2026*

- Co-organized an interactive benchmarking program between STAMANARA and Politeknik Negeri Malang (POLINEMA) Dance Club to exchange organizational insights and foster strategic ties.
- Organized interactive activities and games to encourage engagement between organizations
- Prepared and coordinated the event run-down to ensure the program flow was well structured

Finance & Event Division | Community Service Program *Oct 2025 – Dec 2025*

- Co-organized a community service initiative with the BINUS Modeling Club for SMPN 3 Malang students.
- Managed event budgeting, expense tracking, and resource allocation, and coordinated rundowns, rehearsal schedules, and consumption logistics for the final performance showcase.

Public Relations Division | TKU STAMANARA *May 2025 – Jul 2025*

- Administered participant registration and attendance records for members and alumni, maintaining structured data to support participation tracking and event administration.
- Coordinated participant communications and attendance confirmations, providing updated registration information to support event planning and committee coordination.

BINUS University | Freshmen Leader B29

Freshmen Leader B29 *Jul 2025 – Sep 2025*

- Mentored and supported 45 incoming Computer Science freshmen, facilitating their transition to university life by providing academic guidance, campus orientation, and ongoing peer support.
- Coordinated with orientation leadership to execute orientation schedules seamlessly, achieving a **perfect KPI score of 4.00/4.00** and earning a BINUS Certificate of Appreciation.

Teach For Indonesia Student Community

Class Coordinator | ICSC Community Project *Mar 2025 – Jun 2025*

- Led a student team supporting an economic empowerment initiative for a local herbal medicine UMKM (CV Caraka Abadi), delegating core responsibilities and monitoring execution progress.
- Managed operational schedules, coordinated live class workflows, and authored final progress reports to ensure structured project documentation and successful delivery.
- Authored comprehensive project progress and evaluation reports, documenting operational key results and submission deliverables for stakeholders.

Friends Care Community (FCC)

Master of Ceremony | BINUS Festival Mental Health Workshop *May 2025*

- Served as Master of Ceremony (MC) for the BIFEST Mental Health Workshop, hosting speaker sessions, maintaining audience engagement, and driving smooth event transitions.
- Partnered with the organizing committee to manage event rundowns, coordinate speaker timelines, and ensure seamless operational execution under live workshop conditions.

TECHNICAL SKILLS

Programming & Web Languages: Python, Java, SQL, C, HTML/CSS.

AI & Machine Learning: Deep Learning, Computer Vision, Machine Learning, Exploratory Data Analysis (EDA), Predictive Modeling.

Frameworks & Libraries: Scikit-Learn, Streamlit, Pandas, NumPy.

Cloud & Business Intelligence: Amazon Web Services (AWS), Dashboard Creation, Data Storytelling, Tableau (Current Learning)

Developer Tools & Environments: VS Code, Git/GitHub, Google Colab, Jupyter Notebook, RStudio

Languages: Indonesian (Native), English (Proficient)

LICENSES & CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900) *Mar 2026*

Microsoft | Credential ID: 11154112181489/GreatNusa/III/2026

Front-End Development Certification (High Distinction) *Aug 2025*

Bina Nusantara Computer Club (BNCC) | Credential ID: 052/LnT/III/MEMBER/BNCC/MLG/XXXV/08.2025